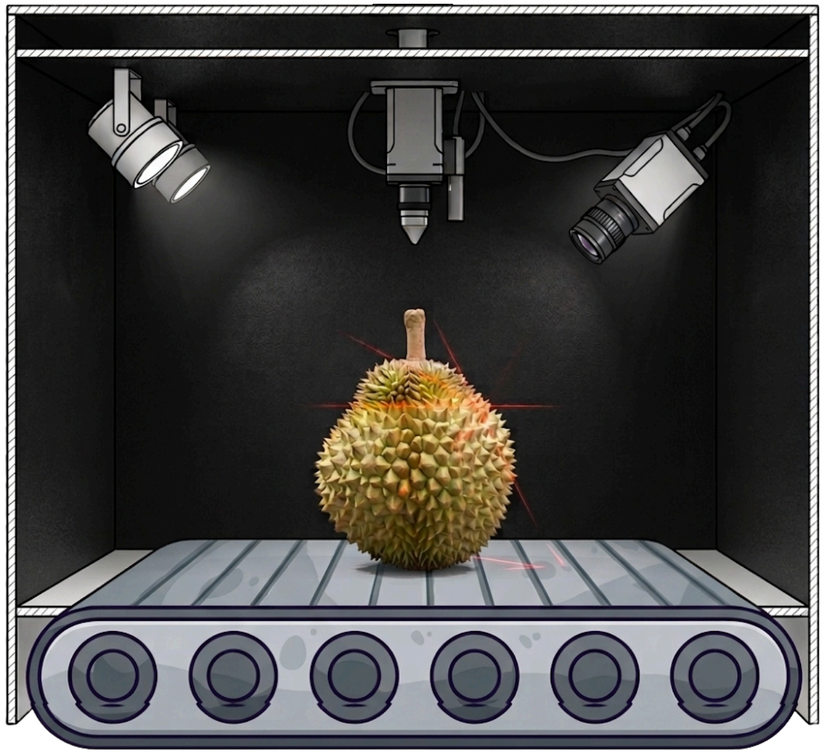
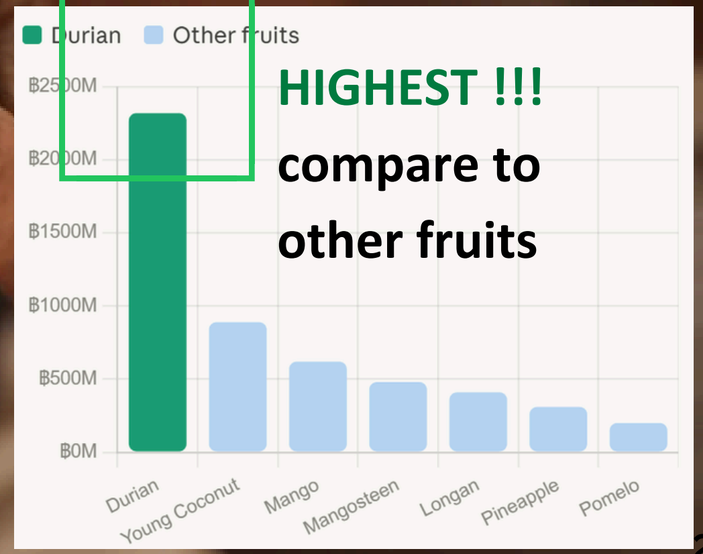
D U R I A N S
C L A S S I F I C A T I O N
S O L U T I O N

TADASHI



AI-Powered, Non-Destructive Durian Quality Classification

Durians export value



HIGHEST !!!

compare to other fruits

MARKET OPPORTUNITIES

IN APRIL 2025

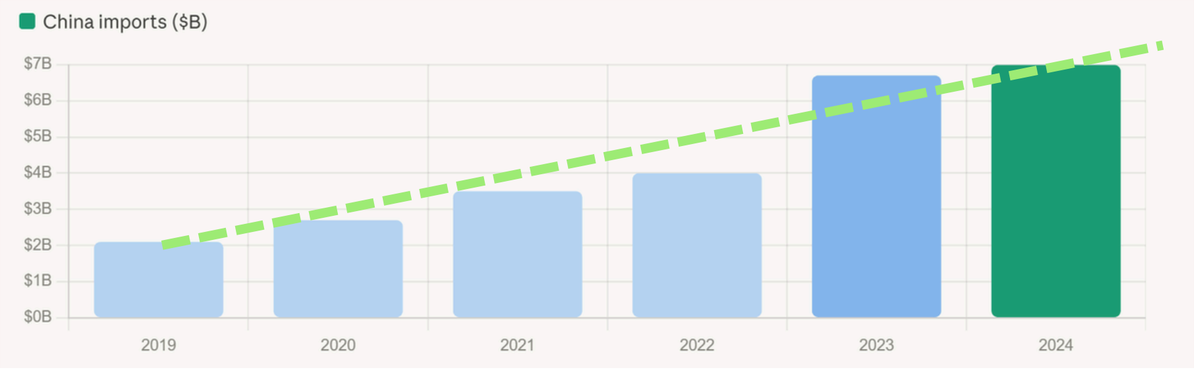
“ the EU granted **approval** for **the import of durian from Thailand** boost demand and increase market size in **Europe** “

New market IN **EUROPE**

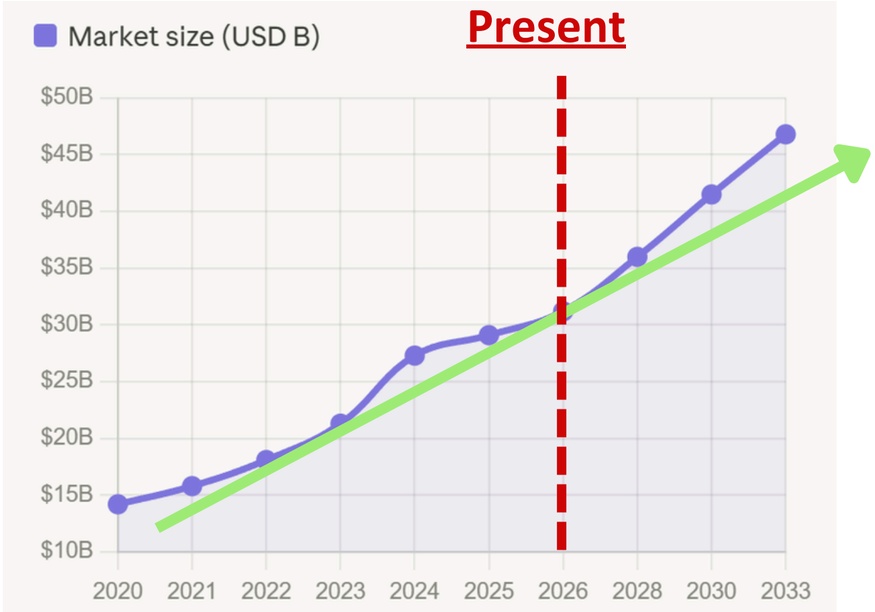
+

The demand for imported durians in **CHINA** has **RISEN SHARPLY**

=



CHINA’S DURIANS IMPORT MARKET

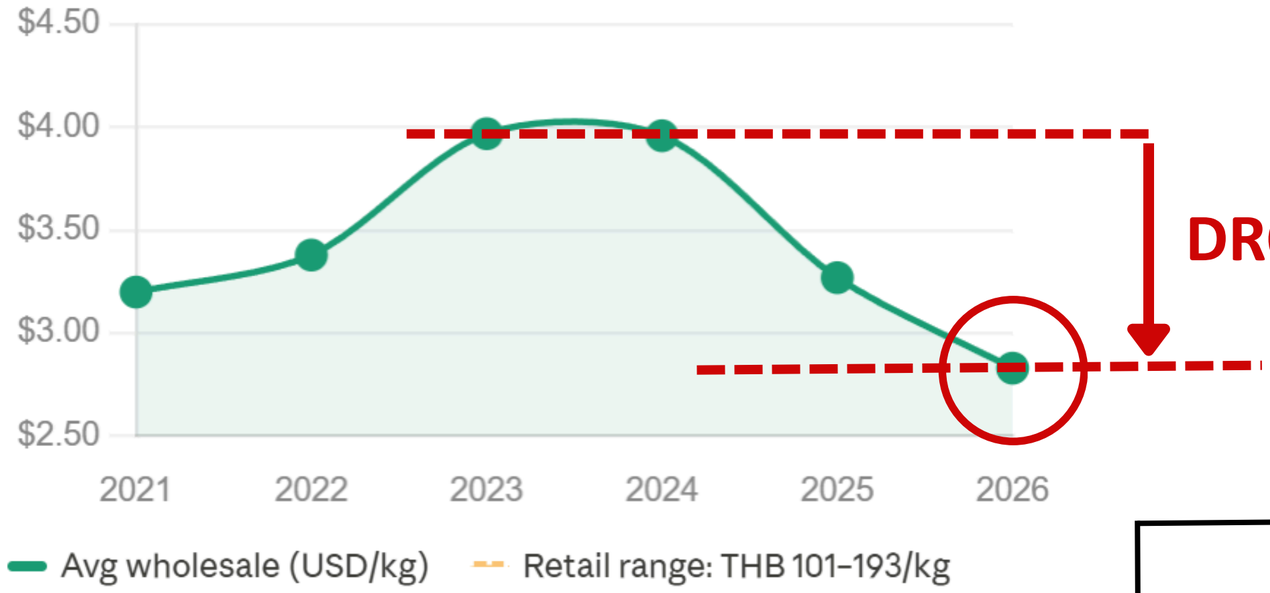


DURIANS MARKET SIZE PROJECTION

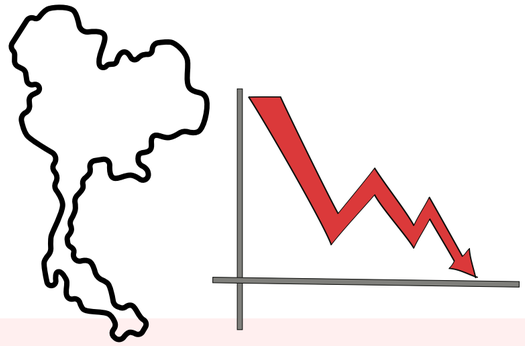
Significant growth

PROBLEM AND PAINPOINTS

1) AVG Durians prices (USD/KG)



DROP 30% OVER 2 YEARS



PRESENT

2) Thai durians market share in China

68% (2023) → **57% (2024)**

Due to : Quality issues



"The average durians value per KG sold from Thailand has decreased, and the overall market value of durian sales in China has also declined"

TARGET CUSTOMER & MARKET SEGMENTATION

Target	Pain point	Willingness to pay	Product Fit	Priority
Lhong (Exporters)	**High** - risk of export rejection (DM < 32%), inconsistent grading, and need labor	**High** - business loss from bad quality is significant	**Very High** - full system (CV + NIR + conveyor) solves automation and accuracy	**High**
Supermarket	**Medium** - cannot guarantee ready to eat or premium quality, customer dissatisfaction	**High** - willing to pay for premium products	High - Black box gives ripeness % for better selling and identification	**Medium**
Wholesale Hubs (Talaad Thai, Si Mum Muang)	**High** - large volume handling, inconsistent sorting quality, and a need for faster operations	**Medium** - cost-sensitive but value efficiency improvements	**Medium** - Black box system is useful, but budget constraints limit adoption	**Medium**
Farmers	**Medium** - want to sell for a better price but not strict in quality	**Low** - limited budget	**Medium** - Black box is useful but too expensive and complex, farmers prefer traditional method	**Low**
Local Vendors (Street carts, Wet markets)	**Medium** - want to avoid selling bad durian but rely on experience	**Low** - very price-sensitive, small scale	**Medium** - Black box is useful, but cost is a barrier	**Low**
Customer	**Low** - occasional uncertainty when buying durian	**Low** - unlikely to buy the device	**Low** - not practical for daily use because it is specifically for durian	**Low**

First Phase

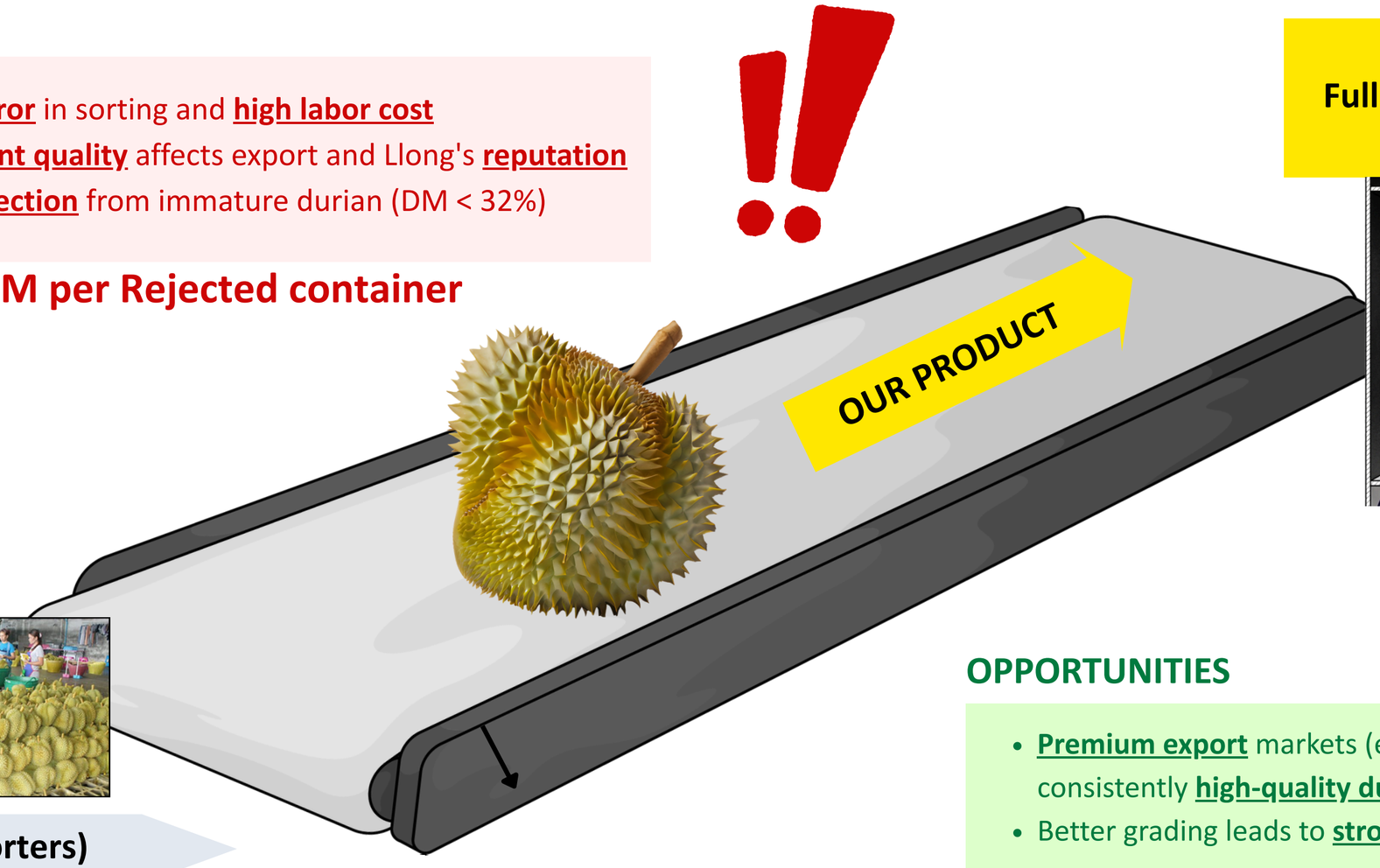
Possible Expansion

CUSTOMER FOCUS: LHONG

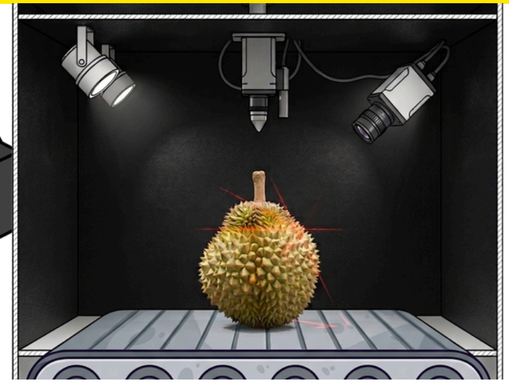
PAIN POINTS

- **Human error** in sorting and **high labor cost**
- **Inconsistent quality** affects export and Llong's **reputation**
- **Export rejection** from immature durian (DM < 32%)

Loss ฿5M per Rejected container



TADASHI
Full automated sorting system
(CV + NIR + conveyor)



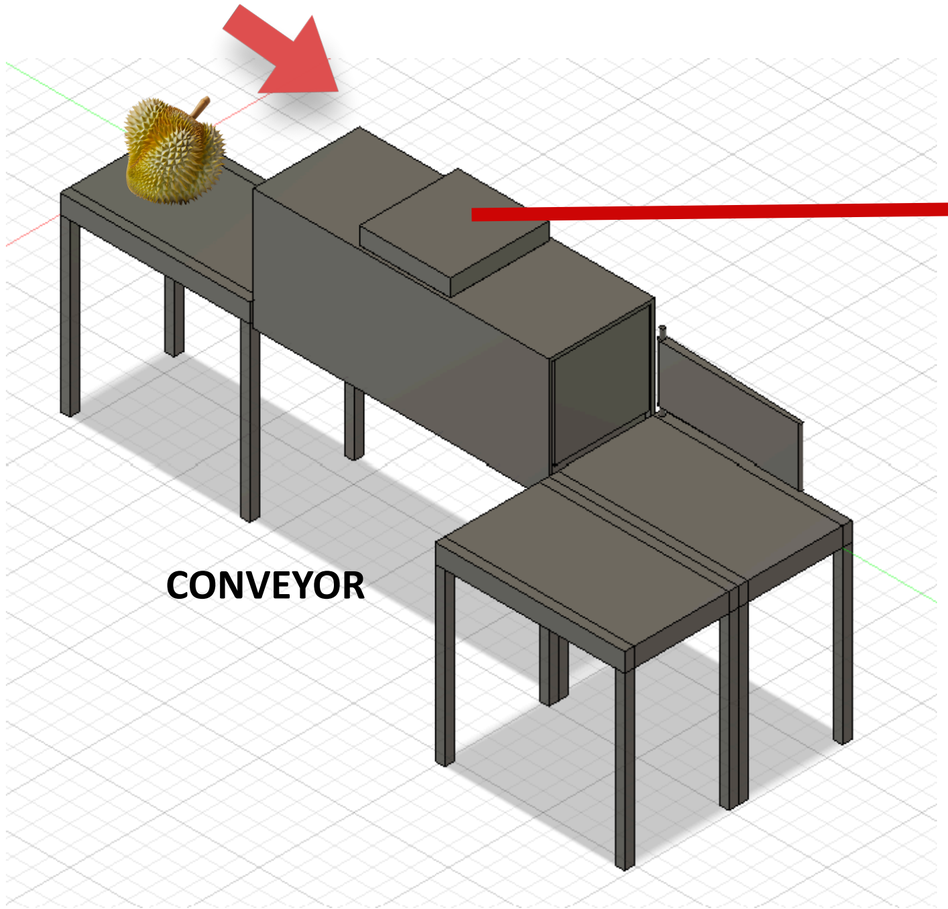


Lhong (Exporters)

Large-scale durian sorting facilities which focus on export quality control and consistency

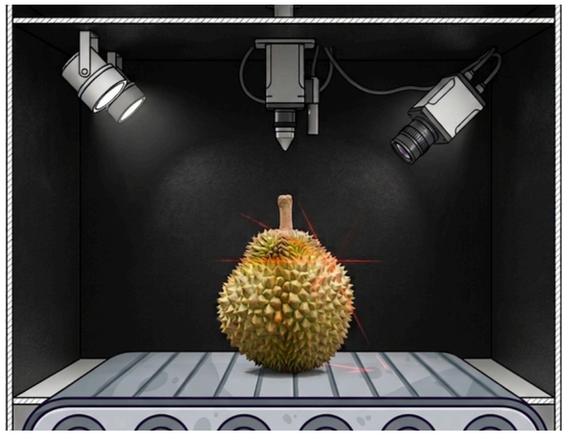
OPPORTUNITIES

- **Premium export** markets (e.g., China) **demand** consistently **high-quality durian**
- Better grading leads to **stronger brand trust**



Introducing TADASHI

Tadashi NiCv-Dr-Sorter evo Gen1



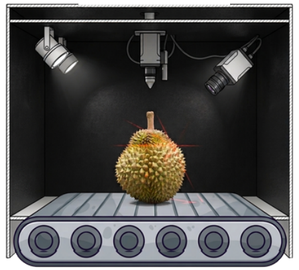
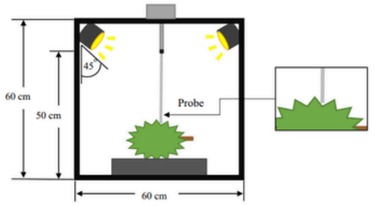
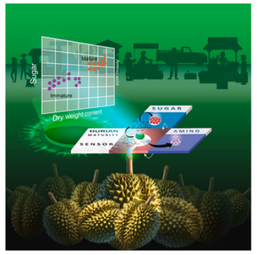
Full automated sorting system
(CV + NIR + conveyor)

MAIN FUNCTIONS

1. Classify the maturity and defects of the durians
2. Automatic system for separating the “high-quality” durians
3. Display current durian classification stats
4. Be able to export logged CSV data for further analysis

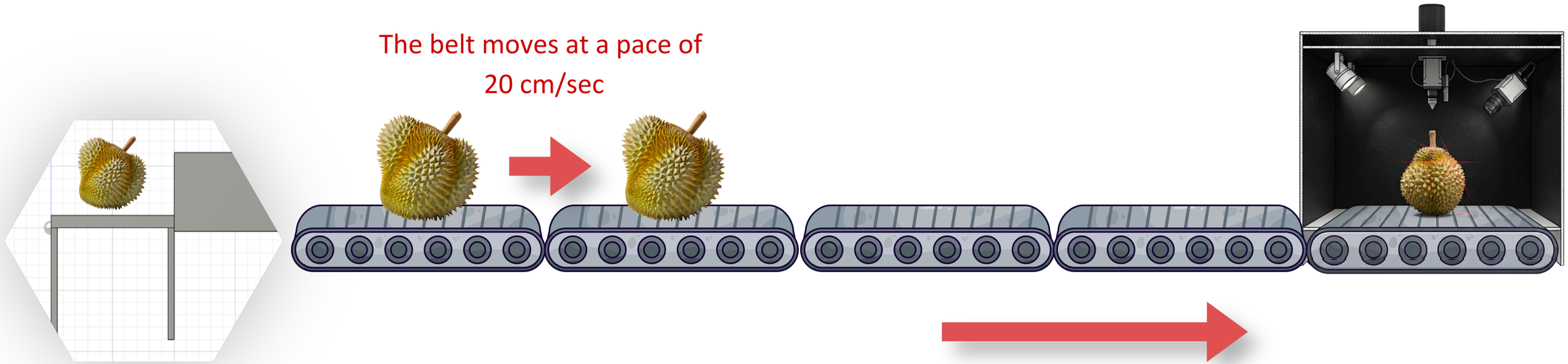
WHY TADASHI ?

Competitor Analysis

	 TADASHI	 MANUAL	 ARDA CT-SCAN	 LWNIR CONVEYER	 µPAD PAPER SENSOR
TYPE	NIR + CV/CNN	HUMAN	X-RAY	NIR	CHEMICAL
SPEED	FAST	MEDIUM	FAST	FAST	SLOW
OUTPUT	INTERNAL EXTERNAL	INTERNAL EXTERNAL	INTERNAL	INTERNAL	INTERNAL
ACCURACY	95%	65%	95%	85%	90%
 COST	MEDIUM 240k	LOW	HIGH 2 M	HIGH 500k-1M	LOW

PRODUCT OPERATION

The belt moves at a pace of
20 cm/sec



Conveyor belt lead durian into a black box (Radiation insulated box)
to be analyzed using NIR and CV.

TADASHI
Upto 1200 durians/hour

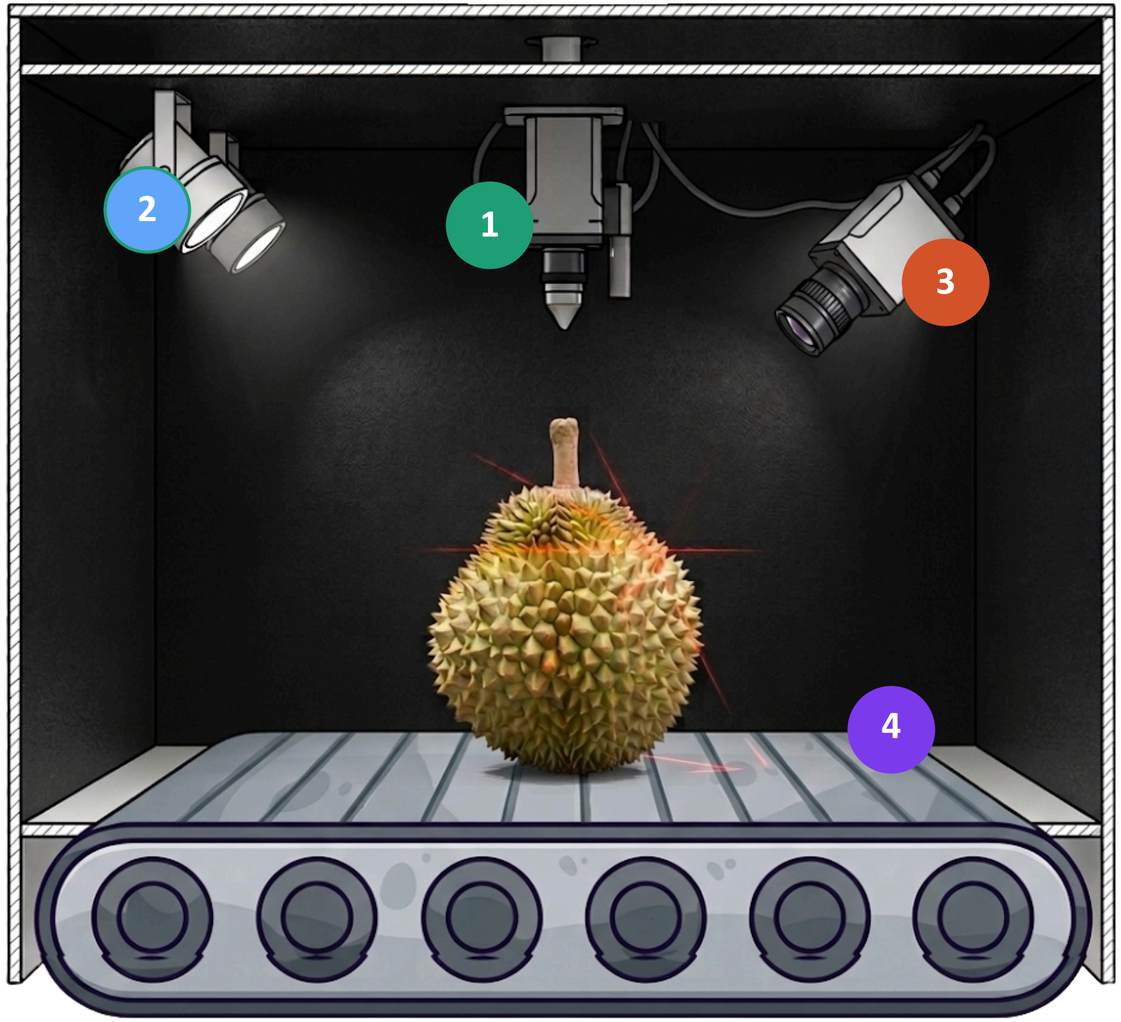
VS

WORKER
360 durians/hour

THEN

The system will then seperate the bad durian from the good ones
using a pusher, pushing it to another line of conveyor belt.

TECHNOLOGY



1 NIR spectrometer — **-> PLSR and LDA model**

Shortwave NIR probe (900-1700 nm) measures absorption to predict dry matter % + Brix and classify maturity

2 Controlled lighting — **-> CV camera feed**

Dark enclosure + halogent light to eliminate ambient noise for consistent CV imaging

3 CV camera — **-> DuNet-12**

Machine vision unit feeds DuNet-12 for defect detection and surface classification

4 Conveyor belt — **-> Downstream gate**

Controls scan timing and triggers accept/reject gate at DM >= 32% threshold

Sensor → **Pre-process** → **Inference** → **Fusion** → **Output**

On-device inference — offline, no cloud latency, works at farm gate

TECHNOLOGY

95%

OVERALL CLASSIFICATION ACCURACY — CV + NIR COMBINED

STEP 1
Imaging
Computer Vision
High-res multi-angle scan on conveyor

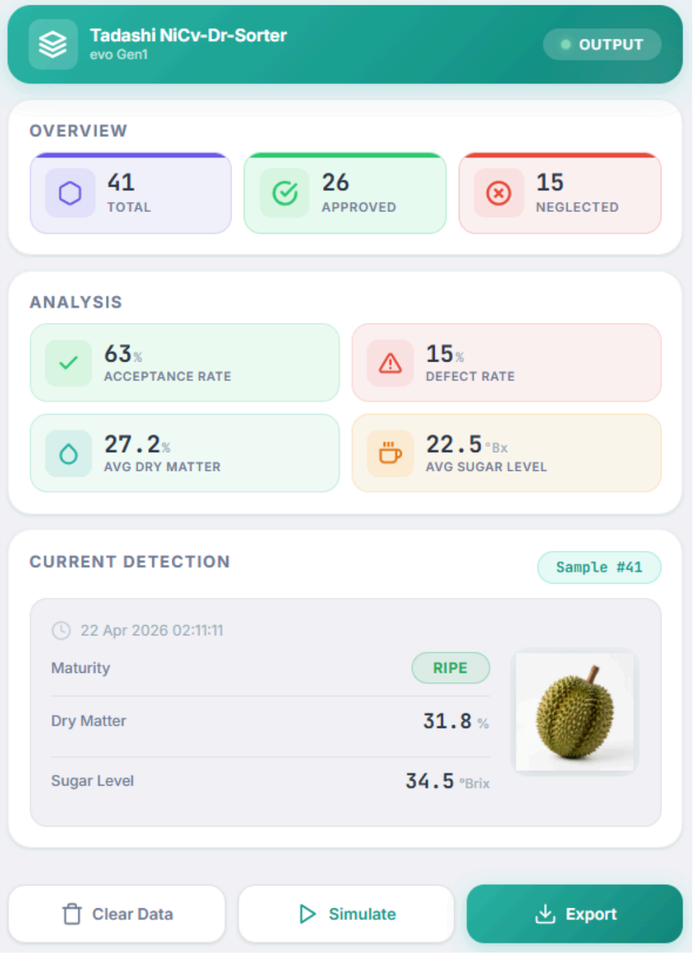
STEP 2
Spectroscopy
SWNIR sensor
Near-infrared absorbance spectrum capture

STEP 3
Inference
DuNet-12 model
Deep neural net trained on durian datasets

OUTPUT
Decision
Real-time grade
Accept / reject + dry matter & sugar estimate

ALGORITHMS

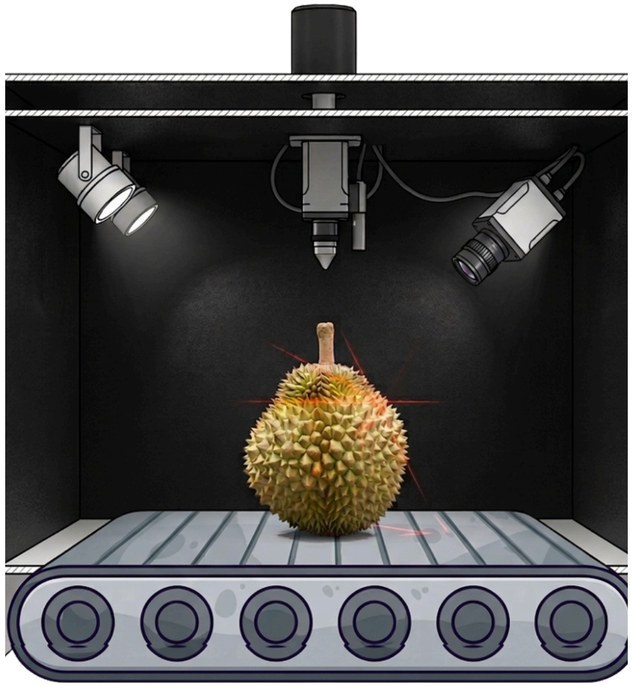
LDA	Maturity classification
PLSR	Dry matter % prediction
DuNet-12	Visual Classification



BUSINESS MODEL

Current cost (COGS)

120,000 THB



B2B

240,000

THB

PER PIECE

FOR LHONG



GROSS MARGIN

100%

SCALABILITY AND MARKETING

JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEPT	OCT	NOV	DEC

STARTING ADVERTISING

GETTING FIRST CUSTOMER

DURIANS EXPORT PEAK SEASON

EXPANSION

CAMPAIGN 1

ADS toward specialized communities like "Durian Traders Thailand" or "Lhong Chanthaburi Owners"

CAMPAIGN 2

Trade Fair Presence

Focus on high-impact events like AGRITECHNICA ASIA or THAIFEX – Anuga Asia to connect manager

BEGIN WITH

PILOT PROJECT

Chanthaburi

need a flagship success story to overcome the skepticism of traditional traders

RESPECTIVELY

1) Follow the harvest **south** = **Chumphon and Yala**
to keep your sales and service teams active during the Eastern off-season

2) Cross-Border Expansion = **Vietnam**
Vietnam is the fastest-growing competitor, with durian exports to China surging to $3.3 billion in 2024.

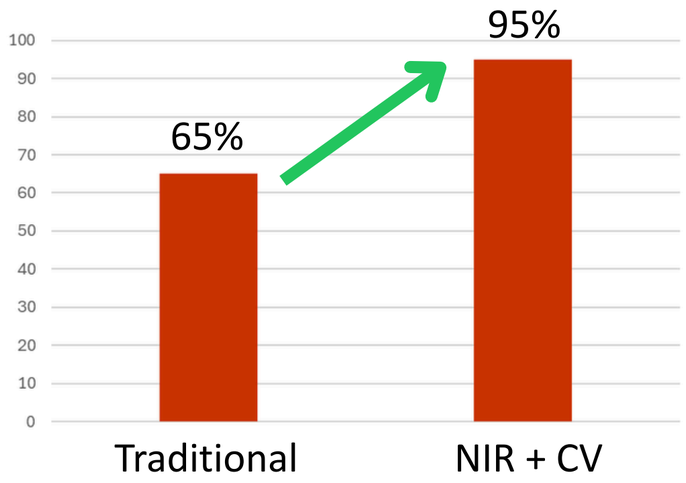
3) Other customer segment = **Supermarket & Wholesale hubs**

Introducing "handheld device"
NOT Black box !



CUSTOMER IMPACT

HIGH ACCURACY FROM “TADASHI”



GUARANTEE EXPORT QUALITY

Ensure ripeness

DM ≥ 32%

No export defects

NO HUMAN LIMITATIONS

Capable of operating **24 hours** without getting tired or sick	compared to	workers working **8 hours** per day
Sort Durian at much faster rate at **3 second** per unit	compared to	**10 second** per unit

Traditional method

STRONGER TRUST AND REPUTATION

Prevent bad batches from exporting → Reinforcing **buyer trust** and long-term **relationships**

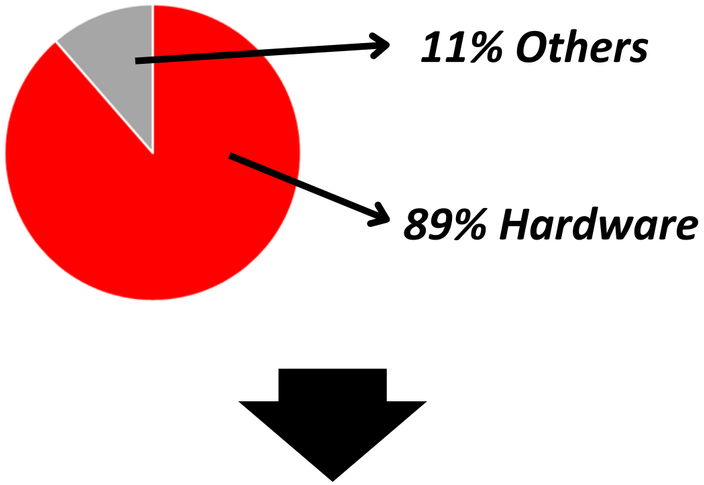
VALUE

Transforming manual export grading into data-driven device to ensure consistent export quality and eliminate export rejection.

FINANCIAL IMPACT

Cost & Pricing

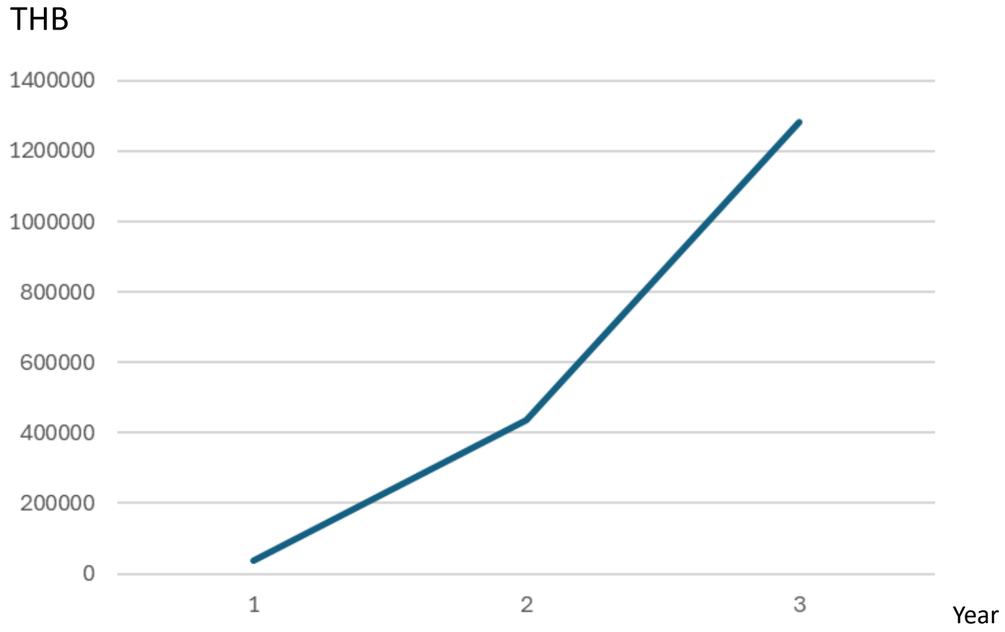
COGS = 122,900 THB



Unit Price = 240,000 THB/Unit

Maintenance Price = 6,145 THB/Year

Revenue Forecast



Break even in **2.4 years**

Key Driver

- Durian generates **$3.7+** billion export value annually
- Thai durian exports grew **+181%** in **5 years**
- Production continues to rise (**~10% YoY** increase)

Members



Thavinthorn Ngaoprommin

Faculty of Mechanical Engineering



Sorrawit Sramoon

Faculty of Commerce and Accountancy



Siraphop Kamhangkhumphol

Faculty of Engineering in Robotics and Artificial Intelligence



Worranai Nagasawa

Faculty of Information and Communication Engineering



Thanavit Wongsoonthorn

Faculty of Electrical Engineering